AAE 33900 Aerospace Propulsion

(3 credits)

COURSE INFORMATION/DESCRIPTION:

This course provides an introduction to and survey of air-breathing and rocket propulsion systems. It fulfills the Propulsion requirement for AAE students who are not concentrating in Propulsion. Students review the first and second laws of thermodynamics; control-volume analysis and the conservation equations of mass, momentum, and energy; the Brayton cycle; and the derivation and application of the rocket equation. The course also provides a foundational background in one-dimensional compressible internal flows, including effects of area change, heat addition, friction, and shocks. Additional topics include cycle and performance analysis of ramjets, turbojets, and turbofans; propellers; and reacting flows and chemical equilibrium. Students are introduced to the major components of air-breathing engines and rockets, such as inlets, turbomachinery, propellers, combustors, and nozzles.

LEARNING RESOURCES AND REFERNCE TEXT

PURDUE BRIGHTSPACE is used to organize the course materials such as announcements, course notes, homework assignments, grades, and solutions. It is important to regularly check Brightspace for the latest update. Lecture notes will be provided before each lecture in Brightspace.

REQUIRED TEXTBOOK none but Mechanics and Thermodynamics of Propulsion, 2nd Edition by Hill and Peterson (H&P) is recommended.

ASSIGNMENTS:

Homework

Weekly homework is assigned. All homework assignments are intended to be completed independently. While seeking assistance from the instructor, TA, or fellow students is encouraged when necessary, the work submitted should reflect one's individual effort.

Exams

There are two mid-exams and one final for the course.

COURSE LEARNING OBJECTIVES:

On completing his course, the student shall be able to:

1. Demonstrate a knowledge of conventional propulsion system types, their form, the functions of their components, and the applications for which they are suited.
2. Calculate the properties of 1D compressible flow with area change, energy exchange (heat and work), friction, and shocks.
3. Apply laws of conservation and the 2nd Law of thermodynamics to calculate thrust and specific impulse.
4. Calculate performance and efficiency of ramjets, turbojets, turbofans, turboprops and rockets and their components including inlets, compressors and pumps, combustors, turbines, and nozzles.
5. Derive propulsion requirements from mission requirements and perform top-level sizing calculations of ramjets, gas turbine engines, liquid rockets, and solid rockets.

EVALUATION:

Course Grade

Mastery of the course content will be assessed through a combination of homework assignments and exams. Student performance on each is graded according to a rubric, yielding a numerical score. The weighted scores are added to form the basis of a letter grade assigned at the end of the semester.

Assignments	Due	Points
Homework	weekly	30
Midterm exams (2)	TBD	40
Final Exam	TBD	30
		Total: 100

Grade Scale

Letter grades are assigned at the end of the semester. The total points earned (TPE) will be calculated for each student at the end of the semester. The mean value and standard deviation of the class TPE distribution will be used as a starting point for assigning letter grades. Notionally, students with TPE above the mean value will receive letter grades from A+ to B-, and students with TPE below the mean value will receive letter grades between C+ and F. Students with TPE greater than the mean plus one standard deviation will receive letter grades of A- and above, and students with TPE less than the mean minus one standard deviation will receive letter grades of D+ and below. Other data from the TPE distribution will be used to make fine adjustments to the final letter grades.

The equivalency between TPE and letter grade varies between offerings and cannot be accurately predicted. In previous years, the equivalency between numerical (%) and letter grades has roughly been:

A: 85 - 100
B: 75 - 84
C: 65 - 74
D: 50 - 64
F: 49 or below